Minimal sign issue in the shortcut-to-absorbing-site calculation

Note on the first hard error in Eqs. (30)–(32)

Where I think the issue starts. The step from the probability-conserving shortcut formula to the absorbing case is made by replacing the ring propagator $\widetilde{P}$ with the absorbing propagator $\widetilde{W}$. This replacement is not valid once the shortcut endpoint is also the absorbing site.

In the manuscript, the absorbing site is then set equal to the shortcut endpoint:

$$m = v, \qquad \widetilde{W}_v(n, z) = 0.$$

At this point, the shortcut $u \to v$ is not an additional transient propagation channel. Since v is absorbing, a walker that takes the shortcut has left the transient state space.

Let

$$\lambda = \beta(1 - q),$$

and let P_0 denote the transition matrix on the non-absorbing states for the ring with the absorbing target v already removed. In transient space, the shortcut to v is therefore a killing term at u:

$$\boxed{P = P_0 - \lambda |u\rangle\langle u|.}$$

Consequently

$$I - zP = I - zP_0 + z\lambda |u\rangle\langle u|.$$

Applying Sherman–Morrison gives

$$(I - zP)^{-1} = (I - zP_0)^{-1} - \frac{z\lambda (I - zP_0)^{-1}|u\rangle\langle u|(I - zP_0)^{-1}}{1 + z\lambda \langle u|(I - zP_0)^{-1}|u\rangle}.$$

Corrected form of Eq. (31).

$$\boxed{\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1 - q)\widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z)}{1 + z\beta(1 - q)\widetilde{W}_u(u, z)}.}$$

The correction term is negative and the denominator has a plus sign.

The sign is forced by the meaning of the absorbing boundary: adding a direct shortcut into the absorbing target removes paths from the transient propagator. Equivalently, to first order in λ the transient occupation should decrease:

$$\widetilde{S} = \widetilde{W} - z\lambda\, \widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z) + O(\lambda^2).$$

The formula with a positive correction and denominator $1 - z\lambda\widetilde{W}_u(u, z)$ would instead increase transient occupation, which corresponds to adding a propagation channel inside the transient state space rather than adding absorption at v.

The first-passage generating function then inherits the corrected sign:

$$\boxed{\widetilde{F}_{n_0}(v, z) = \widetilde{\mathcal{F}}_{n_0}(v, z) + \frac{z\beta(1 - q)\widetilde{W}_{n_0}(u, z)\left[1 - \widetilde{\mathcal{F}}_u(v, z)\right]}{1 + z\beta(1 - q)\widetilde{W}_u(u, z)}.}$$

Thus the later auxiliary denominator should be

$$\frac{\beta(1-q)}{q}\,\frac{1}{1+z\beta(1-q)\widetilde{W}_u(u,z)},$$

not the corresponding expression with $1-z\beta(1-q)\widetilde{W}_u(u,z)$. The later pole and threshold conclusions should therefore be recomputed from this corrected resolvent.